Lesson4 Pulley

Objective: You will be able to

-Describe pulley

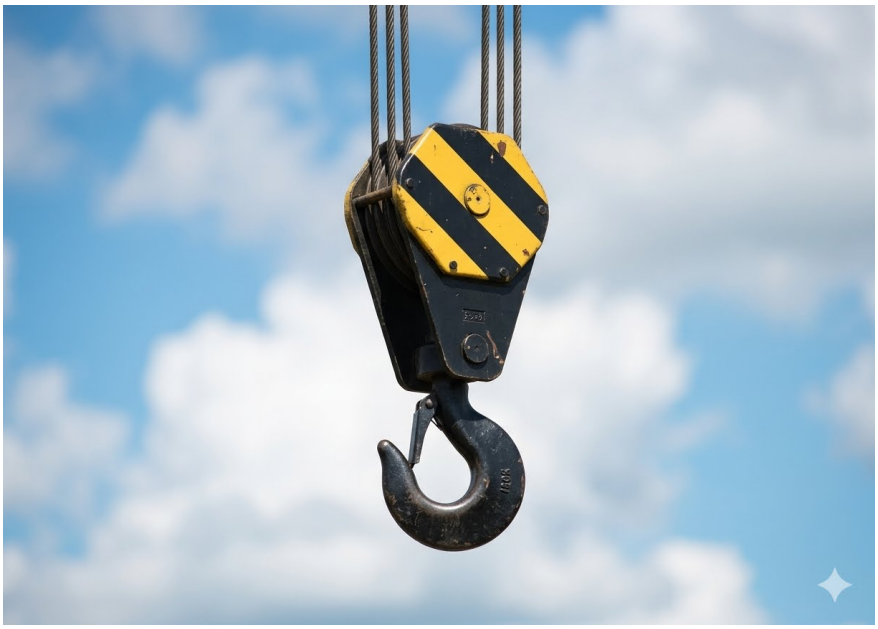
- Calculate mechanical advantage.

-Use pulley in daily life.

1-Pulley

Pulley: A simple machine that changes the direction of a force, often to lift a load. Usually consists of a grooved wheel in which a pulled rope or chain runs. A pulley wheel is a mechanism which helps move or lift objects. Like most wheels, pulley wheels spin or rotate on an axis. The centre of a pulley wheel features a groove. Nested in this groove is a rope, belt or cable.

The man in this image is pulling downwards on a bar, which is attached to a cable. Tracing the cable's path through the machine, it can be seen that the cable passes through the pulley wheels, and its opposite end is connected to the weights at the bottom.



Parts of a Pulley System

Effort – the force the man is applying to the bar

Load – the weight being lifted

Fulcrum – the pivot point of the pulley

Block and Tackle Pulley

This consists of two or more pulleys (fixed and moveable). The block and tackle is capable both of changing the direction and creating a *Mechanical Advantage.*

Fixed pulley: A pulley system in which the pulley is attached to a fixed point and the rope is attached to the object.

Movable pulley: *A pulley system in which the pulley is attached to the object; one end of the rope is attached to a fixed point and the other end of the rope is free.*

2.Mechanical advantage is the ratio of output force(load) to input force(effort).

$$MA = \frac{Load}{Effort} = \frac{F_R}{F_E} = Number\ of\ \text{sup}\ porting\ cables$$

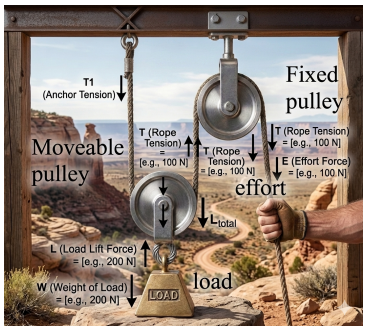
Efficiency of pulley: $efficiency = \frac{useful\ \ work\ \ output}{work\ \ input}$

Or $$Eff = \frac{W_0}{W_{in}} = \frac{F_R \times d_R}{F_E \times d_E}$$

For real machine (friction), the efficiency is less than 100%.

Practice Problems

1.In the compound pulley pictured below, how many Newtons of force are applied to the load if 200 Newtons of effort force are applied to the pulley?



2.How would you construct a compound pulley with an ideal mechanical advantage of 6?

3.A simple pulley system is shown opposite. A 40N effort is used to move a 80N load. The final effort moves 2 meters and at the same time the load moves 1 meter. Find mechanical advantage, velocity ratio, and efficiency.

